

GUILLAUME ALLAIS

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APPOINTMENTS

University of Strathclyde

Chancellor's Fellow (Lecturer)

2023 – ongoing

Glasgow, United Kingdom

- Research (Safe and efficient representation of inductive types in Quantitative Type Theory [2, 3]; Compile-time partial evaluation via model constructions [1]; Sound and complete proof automation for algebraic theories [4])
- Teaching (Functional & Advanced functional programming, Logic gates and circuits)
- Knowledge exchange (Scottish PL summer school)

University of St Andrews

Research Fellow

2020 – 2023

St Andrews, United Kingdom

- Core developer of Idris 2, a bootstrapped research compiler for a dependently typed programming language
- Collaborator on Frex [9, 4], a proof automation framework
- Core developer of TypOS [6], a language of concurrent actors to build typecheckers

University of Strathclyde

Postdoctoral Researcher

2019

Glasgow, United Kingdom

- Core developer of Agda, a research compiler for a dependently typed programming language
- Core developer of Syrup, a home grown language to teach logic circuits to undergraduates

Radboud Universiteit

Postdoctoral Researcher

2016 – 2018

Nijmegen, The Netherlands

- Formalising in Agda a resource-aware type system based on linear logic [15]

EDUCATION

University of Strathclyde

PhD in Computer Science

2022

Glasgow, United Kingdom

- Generic programming and proving framework for syntaxes with binding [8]
- Core developer of Agda, a research compiler for a dependently typed programming language

Parisian Research Master in Computer Science (MPRI)

MSc in Computer Science

2011

Paris, France

- Type theory and interactive theorem proving
- Research internship with René Thiemann on certifying termination of term rewrite systems in Isabelle [19]

École Normale Supérieure de Lyon

BSc in Computer Science

2009

Lyon, France

- Research internship with Yves Bertot on real analysis in Coq [17]

TEACHING

Functional Programming, University of Strathclyde

Co-delivering lectures with Bob Atkey, designing and marking coursework and examinations

2024–ongoing

Computer Systems and Architecture, University of Strathclyde

Co-delivering lectures with Alasdair Lambert, core developer of the educational circuit-generation programming language used in the course

2024–ongoing

Advanced Functional Programming, University of Strathclyde 2023–ongoing
Co-delivering lectures with Dr Conor Mc Bride and Dr Fredrik Nordvall Forsberg, writing lecture notes, designing and marking coursework and examinations

Computer Systems and Architecture, University of Strathclyde 2019
Delivering lectures, designing and marking coursework and examinations

ORGANISATIONAL DUTIES

International Conference on Types for Proofs and Programs, University of Strathclyde 2025
Local co-organiser

Scottish Programming Languages and Verification Summer School, University of Strathclyde 2024
Lead organiser

MSPrior seminar, University of Strathclyde — *Organiser* 2023

PROGRAM COMMITTEE MEMBERSHIP

GPCE, International Conference on Generative Programming, 2025

ICFP, International Conference on Functional Programming, 2025

TYPES, International Conference on Types for Proofs and Programs, 2025

PEPM, Workshop on Partial Evaluation and Program Manipulation 2025 (Co-chair), 2021, 2020

TyDe, Workshop on Type-driven Development 2024, 2018

TFP, Trends in Functional Programming 2022

REFEREEING

Journals	Journal of Functional Programming (2024, 2019)
Conferences	Certified Programming and Proofs (2025), Principles of Programming Languages (2023), European Symposium on Programming (2022), Computer Science Logic (2022), Mathematically Structured Functional Programming (2014), Foundations of Software Science and Computation Structures (2022, 2013)

PROFESSIONAL ACTIVITIES AND VISITS

Guest, IFIP Working Group 2.11 2024
University of Edinburgh (United Kingdom)

Invited Speaker, School “Formal Mathematics” 2024
Hausdorff Institute for Mathematics (Bonn, Germany)

Invited Lecturer, Scottish Programming Languages Summer School 2023
University of St Andrews (United Kingdom)

Invited Speaker, Workshop Compiler Coding Dojo 2019
Co-located with <Programming> (Genoa, Italy)

INVITED SEMINAR PRESENTATIONS

PLUG, University of Glasgow, UK 2023-11-08
Staging by Evaluation

Séminaire Gallinette, Nantes Université, France 2023-05-22
Seamless, Correct, and Generic Programming over Serialised Data

RESEARCH SOFTWARE DEVELOPMENT

TypOS , Core Developer <i>An actor-based concurrent language for unification-based elaborators and typecheckers</i>	2021 – ongoing
Idris 2 , Core Developer <i>Bootstrapped compiler for Quantitative Type Theory</i>	2020 – ongoing
Agda Standard Library , Core Developer <i>Certified data structures, discrete mathematics proofs, and automated solvers for the Agda programming language</i>	2017 – ongoing
Agda , Core Developer <i>Compiler for Martin-Löf Type Theory</i>	2015 – ongoing
COQTAIL , Core Developer & Administrator <i>Real and complex analysis in Coq</i>	2010 – 2012

PUBLICATIONS

- [1] Guillaume Allais. “Scoped and Typed Staging by Evaluation”. In: *Proceedings of the 2024 ACM SIGPLAN International Workshop on Partial Evaluation and Program Manipulation, PEPM 2024, London, UK, 16 January 2024*. Ed. by Gabriele Keller and Meng Wang. ACM, 2024, pp. 83–93. DOI: 10.1145/3635800.3636964. URL: <https://doi.org/10.1145/3635800.3636964>.
- [2] Guillaume Allais. “Builtin Types Viewed as Inductive Families”. In: *Programming Languages and Systems - 32nd European Symposium on Programming, ESOP 2023, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2023, Paris, France, April 22-27, 2023, Proceedings*. Ed. by Thomas Wies. Vol. 13990. Lecture Notes in Computer Science. Springer, 2023, pp. 113–139. DOI: 10.1007/978-3-031-30044-8_5. URL: https://doi.org/10.1007/978-3-031-30044-8_5.
- [3] Guillaume Allais. “Seamless, Correct, and Generic Programming over Serialised Data”. In: *CoRR abs/2310.13441* (2023). Submitted to JFP. DOI: 10.48550/ARXIV.2310.13441. arXiv: 2310.13441. URL: <https://doi.org/10.48550/arXiv.2310.13441>.
- [4] Guillaume Allais, Edwin C. Brady, Nathan Corbyn, Ohad Kammar, and Jeremy Yallop. “Frex: dependently-typed algebraic simplification”. In: *CoRR abs/2306.15375* (2023). Draft. DOI: 10.48550/ARXIV.2306.15375. arXiv: 2306.15375. URL: <https://doi.org/10.48550/arXiv.2306.15375>.
- [5] Jan de Muijnck-Hughes, Guillaume Allais, and Edwin C. Brady. “Type Theory as a Language Workbench”. In: *Eelco Visser Commemorative Symposium, EVCS 2023, April 5, 2023, Delft, The Netherlands*. Ed. by Ralf Lämmel, Peter D. Mosses, and Friedrich Steimann. Vol. 109. OASICS. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023, 9:1–9:13. DOI: 10.4230/OASICS.EVCS.2023.9. URL: <https://doi.org/10.4230/OASICS.EVCS.2023.9>.
- [6] Guillaume Allais, Malin Altenmüller, Conor McBride, Georgi Nakov, Fredrik Nordvall Forsberg, and Craig Roy. “TypOS: An Operating System for Typechecking Actors”. In: *28th International Conference on Types for Proofs and Programs, TYPES 2022, June 20-25, 2022, Nantes, France*. 2022.
- [7] Guillaume Allais, Matthew L. Daggett. “Using Dependent Types at Scale: Maintaining the Agda Standard Library”. In: *Workshop on the Implementation of Type Systems, WITS@POPL 2022, Philadelphia, Pennsylvania, USA, January 22, 2022*. 2022.
- [8] Guillaume Allais, Robert Atkey, James Chapman, Conor McBride, and James McKinna. “A type- and scope-safe universe of syntaxes with binding: their semantics and proofs”. In: *Journal of Functional Programming* 31 (2021). Extended version, e22. DOI: 10.1017/S0956796820000076. URL: <https://doi.org/10.1017/S0956796820000076>.
- [9] Guillaume Allais, Edwin Brady, Ohad Kammar, and Jeremy Yallop. “Frex: indexing modulo equations with free extensions”. In: *The workshop on Type-Driven Development, TyDe 2020, August 23, 2020*. 2020.
- [10] Andreas Abel, Guillaume Allais, Aliya Hameer, Brigitte Pientka, Alberto Momigliano, Steven Schäfer, and Kathrin Stark. “POPLMark reloaded: Mechanizing proofs by logical relations”. In: *J. Funct. Program.* 29 (2019), e19. DOI: 10.1017/S0956796819000170. URL: <https://doi.org/10.1017/S0956796819000170>.
- [11] Guillaume Allais. “Generic level polymorphic n-ary functions”. In: *Proceedings of the 4th ACM SIGPLAN International Workshop on Type-Driven Development, TyDe@ICFP 2019, Berlin, Germany, August 18, 2019*. Ed. by David Darais and Jeremy Gibbons. ACM, 2019, pp. 14–26. ISBN: 978-1-4503-6815-5. DOI: 10.1145/3331554.3342604. URL: <https://doi.org/10.1145/3331554.3342604>.
- [12] Guillaume Allais. “Scrap your DSL Boilerplate with a Universe of Syntaxes, their Programs and Proofs”. Invited Talk at CoCoDo’19. 2019.

- [13] Guillaume Allais. “agdarsec – Total Parser Combinators”. In: *Journées Francophones des Langages Applicatifs 2018*. Ed. by Sylvie Boldo and Nicolas Magaud. Banyuls-sur-Mer, France, Jan. 2018, pp. 45–59. URL: <https://hal.inria.fr/hal-01707376>.
- [14] Guillaume Allais, Robert Atkey, James Chapman, Conor McBride, and James McKinna. “A type and scope safe universe of syntaxes with binding: their semantics and proofs”. In: *PACMPL* 2.ICFP (2018), 90:1–90:30. DOI: 10.1145/3236785. URL: <https://doi.org/10.1145/3236785>.
- [15] Guillaume Allais. “Typing with Leftovers - A mechanization of Intuitionistic Multiplicative-Additive Linear Logic”. In: *23rd International Conference on Types for Proofs and Programs, TYPES 2017, May 29-June 1, 2017, Budapest, Hungary*. Ed. by Andreas Abel, Fredrik Nordvall Forsberg, and Ambrus Kaposi. Vol. 104. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum fuer Informatik, 2017, 1:1–1:22. ISBN: 978-3-95977-071-2. DOI: 10.4230/LIPIcs.TYPES.2017.1. URL: <https://doi.org/10.4230/LIPIcs.TYPES.2017.1>.
- [16] Guillaume Allais, James Chapman, Conor McBride, and James McKinna. “Type-and-scope safe programs and their proofs”. In: *Proceedings of the 6th ACM SIGPLAN Conference on Certified Programs and Proofs, CPP 2017, Paris, France, January 16-17, 2017*. Ed. by Yves Bertot and Viktor Vafeiadis. ACM, 2017, pp. 195–207. ISBN: 978-1-4503-4705-1. DOI: 10.1145/3018610.3018613. URL: <https://doi.org/10.1145/3018610.3018613>.
- [17] Yves Bertot and Guillaume Allais. “Views of PI: Definition and computation”. In: *J. Formaliz. Reason.* 7.1 (2014), pp. 105–129. DOI: 10.6092/ISSN.1972-5787/4343. URL: <https://doi.org/10.6092/issn.1972-5787/4343>.
- [18] Guillaume Allais, Conor McBride, and Pierre Boutillier. “New equations for neutral terms: a sound and complete decision procedure, formalized”. In: *Proceedings of the 2013 ACM SIGPLAN workshop on Dependently-typed programming, DTP@ICFP 2013, Boston, Massachusetts, USA, September 24, 2013*. Ed. by Stephanie Weirich. ACM, 2013, pp. 13–24. ISBN: 978-1-4503-2384-0. DOI: 10.1145/2502409.2502411. URL: <https://doi.org/10.1145/2502409.2502411>.
- [19] René Thiemann, Guillaume Allais, and Julian Nagele. “On the Formalization of Termination Techniques based on Multiset Orderings”. In: *23rd International Conference on Rewriting Techniques and Applications (RTA’12), RTA 2012, May 28 - June 2, 2012, Nagoya, Japan*. Ed. by Ashish Tiwari. Vol. 15. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum fuer Informatik, 2012, pp. 339–354. ISBN: 978-3-939897-38-5. DOI: 10.4230/LIPIcs.RTA.2012.339. URL: <https://doi.org/10.4230/LIPIcs.RTA.2012.339>.